STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-BASE-CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 764-780 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 782-795, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 47 of queue_algebraic-stacks-sheaves.json (pb=423, stmt 764-780, proof 782-795).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma equation-pushforward: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 747–751. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} \mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X} & \xrightarrow{g'} & \mathcal{X} \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{Y}' & \xrightarrow{g} & \mathcal{Y} \end{array}$$

be a 2-cartesian diagram of categories fibred in groupoids over S. Then we have a canonical isomorphism

$$g^{-1} f_* \mathcal{F} \longrightarrow f'_* (g')^{-1} \mathcal{F}$$

functorial in the presheaf $\mathcal{F}$ on $\mathcal{X}$.

Proof. Given an object y' of $\mathcal{Y}'$ over V there is an equivalence

$$(Sch/V)_{fppf} \times_{g(y'),\mathcal{Y}} \mathcal{X} = (Sch/V)_{fppf} \times_{y',\mathcal{Y}'} (\mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X})$$

Hence by (equation-pushforward) a bijection $g^{-1} f_* \mathcal{F}(y') \to f'_* (g')^{-1} \mathcal{F}(y')$. We omit the verification that this is compatible with restriction mappings. □

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.